

Find-a-Gene Project

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Alishba Qadri

Q1.

Name: RBP4 protein

Species: Homo sapiens (human)

Accession Number: P02753

Known Function: It carries retinol (vitamin A alcohol) from liver stores, through blood, and into peripheral tissues. It causes vitamin A deficiency if it is blocked and unable to supply retinol to epidermal cells.

Q2.

Blast Method: TBLASTN

Database: Expressed Sequence Tags (est)

Organism: Amphibia (taxid:8292)

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Translated BLAST: tblastn

blastn blastp blastx **tblastn** tblastx

TBLASTN search translated nucleotide databases using a protein query. [more...](#)

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sp|P02753

Query subrange [?](#)

From

To

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Job Title

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Choose Search Set

Database ?

Organism Optional ☐ exclude [Add Organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude Optional ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to Optional ☐ Sequences from type material

Entrez® Query Optional [YouTube](#) [Create custom database](#)

Enter an Entrez query to limit search [?](#)

Search database est using Tblastn (search translated nucleotide databases using a protein query)

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Note: Parameter values that differ from the default are highlighted in yellow and marked with ♦

[+ Algorithm parameters](#)

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Chosen match: Accession CN064263.1, a 731 base pair clone from *Ambystoma tigrinum*. First match of screenshot below.

[< Edit Search](#)[Save Search](#)[Search Summary ▾](#)[How to read this report?](#)[BLAST Help Videos](#)[Back to Traditional Results Page](#)**i** Your search is limited to records include: **Amphibia (taxid:8292)**

Job Title	P02753:RecName: Full=Retinol-binding protein...
RID	Y8H10XGZ014 Search expires on 04-20 10:55 am Download All ▾
Program	TBLASTN Citation ▾
Database	est See details ▾
Query ID	P02753.3
Description	RecName: Full=Retinol-binding protein 4; AltName: Full=P...
Molecule type	amino acid
Query Length	201
Other reports	?

Filter Results

Organism	only top 20 will appear	☐ exclude
Type common name, binomial, taxid or group name		
+ Add organism		
Percent Identity	E value	Query Coverage
to	to	to
		Filter Reset

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Sequences producing significant alignments

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	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	2e-93	69.61%	731	CN064263.1
☒	Salamander_10_I03.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	3e-93	70.00%	735	CN052017.1
☒	Salamander_20_N22.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	3e-93	70.00%	760	CN055612.1
☒	Salamander_2_G02.ab1 AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	4e-93	70.00%	785	CN058168.1
☒	Ag2_p35_I10_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	2e-92	70.00%	913	CN062105.1
☒	Ag2_p36_A19_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	3e-92	70.00%	958	CN062231.1
☒	v3_p1_a3_I3_V3 Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	272	272	88%	5e-92	70.62%	697	CN049128.1
☒	Ag2_p38_K1_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	6e-92	70.00%	1012	CN062855.1
☒	Ag2_p38_2_D23_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence	Ambystoma tigrin...	276	276	90%	1e-91	70.00%	1094	CN062829.1

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Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence

Sequence ID: [CN064263.1](#) Length: 731 Number of Matches: 1Range 1: 115 to 657 [GenBank](#) [Graphics](#)[Next Match ▾](#) [Previous Match ▲](#)

Score	Expect	Method	Identities	Positives	Gaps	Frame
276 bits(707)	2e-93	Compositional matrix adjust.	126/181(70%)	153/181(84%)	0/181(0%)	+1
Query 14	GSGRAERDCRVSSFRVKNFDFKARFSGTWYAMAKKDPEGLFLQDNIVAE					
	S AERDC VS+F V +NFD+ R++GTWYAMAKKDPEGLFL DNIVA+					
Sbjct 115	ASCLAERDCTVSNFPVMQNFDRYRTGTWYAMAKKDPEGLFLLDNIVAK					
Query 74	TAKGRVRLNNWDVCADMVGTFTDTEPAKFKMKYGVASFQKGNDDH					
	TAKGRV + + +CADMVG FTDT+DPAKF M+Y G+ ++L+ G D H					
Sbjct 295	TAKGRVIFGGFQLCADMVGIFTDTPAKFVMQYRGLLAYLENGQDPH					
Query 134	VQYSCRLNLNDGTCADSYSFVFSRDPNGLPPEAQKIVRQRQEELCLARQ					
	+ YSCR LN DGTCADSYSFVFSRDPNGL PEAQ+IVR+RQ+ELCL R+					
Sbjct 475	LHYSRRLNDDGTCADSYSFVFSRDPNGLTPEAQIRVRKQKELCLDRE					
Query 194	G	194				
	G					
Sbjct 655	G	657				

Q3.

A. tigrinum protein GLGCGIGKMDMKACGLLVALGFVASCLAERDCTVSNFPVMQNFDRYRYT-
GTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLATAKGRVVIFGGFQLCADMVGIFT-
DTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYS-
FVFSRDPNGLTPEAQRIVRKRQKELCLDREYRMVAHNGYCEGNKYE

Name: Ambystoma protein

Species: Ambystoma tigrinum (tiger salamander)

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Amphibia; Batrachia;
Caudata; Salamandroidea; Ambystomatidae; Ambystoma

>CN064263.1_1 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
XXXXXXXXXXLTH*GLGCGIGKMDMKACGLLVALGFVASCLAERDCTVSNFPVMQNFD
YRYTGTWYAMAKKDPEGLFLLDNIVAKFTVDDYGRMLATAKGRVVFGGFQLCADMVGIF
TDTDDPAKFVMQYRGLLAYLENGQDPHWVVDTDYDTYALHYSCRRLNDDGTCADSYFVF
SRDPNGLTPEAQIRVRKRQKELCLDREYRMVAHNGYCEGNKYE*DLPKTS*TANPPKINK
TLEX

>CN064263.1_2 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
XXXXXXXXXX*PTEVWAVASARWT*RRVGFWWPASPAPVWQKETARSATSQSCRISTG
TGTQARGMPWPRRIPRGCSFWITLSPSSQWTIMGECSPQPRGVSSFLGVFSYALTWLGYS
RTLMTLPSSSCNTVDSWPTSRMDRTLIGWWTLIMTRTHFITRAAG*MMAPARTATPSSS
PGTPMA*LQRHRESCGRDRKSCVWTGNTGWWPIMDTARETNTSEIFPKPAEQTPPK*TK
LWNX

>CN064263.1_3 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
XXXXXXXXXXFNPLRSGLWHRQDGHGVWASGGPGLRGQLSGRKRLHGQQLPSHAEFRPV
PVHRHVCHGQEGSRGAVPSG*HCRQVHSGRLWENARHSQGACRHFVGFVSMR*HGWDIH
GH**PCQVRHAIPWTPGLPREWTGPSLGGGH*L*HVRTSLLVPQVE**WHLRGQLLLRLL
PGPQWLDSRGTENRAEETERAVSGQGIQDGGP*WILRGKQIRVRSSQNQLNSKPPQNKQN
FGM

>CN064263.1_4 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
FQSFVYFGGVCCSAGFGKISLVFVSLAVSIMGHHPVFPVQTQLFLSLPHDSLCLWSQAIG
VPGEDGEVAVRAGAI IQPAARVMKCVRVIIISVHHPMRVLSILEVGQESTVLHDELGRVI
SVREYPNHVSA*LKTPKNDDTPLGCGEHSPIIVHCELGDNVIQKEQPLGILLGHGIPRAC
VPVPVEILHDWEVADRAVSFCQTAGHEAQGHQKPTRLHVHLADATAQTSVG*XXXXXXXX
XXX

>CN064263.1_5 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
IPKFCLFWGGLLFSWFEDLTRICFPRSIHYGPPSCIPCPDTALSVSSARFSVPLESSHW
GPGRRRRSCPRRCHHSTCGTSNEVRTCHNQCPPNEGVPVHSRGRPGVHGIA*RTWQGH
QCP*ISQPCQRITENPQK*RHAPWLWRAFSHNRPL*TWRCYPEGTAPRDPSPWHTTCL
CTGTGRNSA*LGSC*PCSLFLPDSWPRSPGPPEAHTPSCPSCRCHSPDL SGLXXXXXXXX
XXXX

>CN064263.1_6 Ag2_p47_I4_M13R AG Ambystoma tigrinum tigrinum cDNA, mRNA sequence
HSKVLFILGGFAVQLVLGRSHSYLFPSQYPLWATILYSLSRHSSFCFLRTILCASGVKPL
GSREKTKE*LSAQVPSSFNL RHE**SAYVS*SVSTTQ*GSCPFSR*ARSPRYCMTNLAGS
SVSVNIPTMSAHN*KPPKMTTRPLAVASILP*SSTVNLATMLSRRNPSGSLAMAYHVP
VYRYSKFCMTGKLLTVQSL SARQLATKPRATRSPHAFMSILPMPQPRPQWVKXXXXXX
XXXX

Q4.

A BLASTP search against NR database yielded a top hit result to a protein from *Ambystoma mexicanum* (axolotl).

blastn

blastp

blastx

tblastn

tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. more...

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) ? Clear

>unnamed protein product
GLGCGIGKMDMKACGLLVALGFVASCLAERDCTVSNFPVMQNFDTRYRYGTW
YAMAKKDPEGLFLLDNIV
AKFTVDDYGRMLATAKGRVIFGGFQLCADMVGIFTDTDDPAKFVMQYRGLLAY

Query subrange ?

From

To

Or, upload file

Choose File

No file chosen ?

Job Title

Protein Sequence

Enter a descriptive title for your BLAST search ?

☐ Align two or more sequences ?

Choose Search Set

Database

Non-redundant protein sequences (nr) ?

Organism

Optional

Enter organism name or id--completions will be suggested

☐ exclude

Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. ?

Exclude

Optional

☐ Models (XM/XP)

☐ Non-redundant RefSeq proteins (WP)

☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

☐ PHI-BLAST (Pattern Hit Initiated BLAST)

☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm ?

BLAST

Search database nr using Blastp (protein-protein BLAST)

☐ Show results in a new window

Feedback

The top result is to a protein from *Ambystoma mexicanum* (axolotl).

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	retinol-binding protein 4 [Ambystoma mexicanum]	Ambysto...	413	413	96%	3e-145	99.00%	200	XP_069467453.1
✓	retinol-binding protein 4 [Ascaphus truei]	Ascaphu...	322	322	93%	6e-109	77.84%	195	XP_075468936.1
✓	retinol-binding protein 4 [Microcaecilia unicolor]	Microca...	308	308	94%	2e-103	70.77%	198	XP_030058896.1
✓	retinol-binding protein 4 [Lissotriton helveticus]	Lissotrito...	304	304	96%	5e-102	75.00%	200	XP_078499520.1
✓	retinol-binding protein 4 [Pleurodeles waltl]	Pleurode...	303	303	95%	1e-101	74.24%	199	XP_069095936.1
✓	retinol-binding protein 4 [Pelobates fuscus]	Pelobate...	301	301	93%	6e-101	70.10%	195	XP_063291070.1
✓	retinol-binding protein 4 [Ranitomeya variabilis]	Ranitom...	298	298	93%	2e-99	71.21%	201	XP_077114717.1
✓	retinol-binding protein 4 [Dendrobates tinctorius]	Dendrob...	296	296	93%	7e-99	70.71%	201	XP_073411936.1
✓	retinol-binding protein 4 [Ranitomeya imitator]	Ranitom...	295	295	93%	2e-98	70.20%	201	XP_069609869.1
✓	retinol-binding protein 4 [Rhinoderma darwinii]	Rhinode...	294	294	94%	4e-98	69.70%	198	XP_075697141.1
✓	retinol-binding protein 4 [Anomaloglossus baeobatrachus]	Anomalo...	294	294	94%	5e-98	69.85%	201	XP_075204695.1

In order to confirm my protein was novel, I blasted against my species (*Ambystoma tigrinum*) only. I received no matches.

blastn

blastp

blastx

tblastn

tblastx

Standard Protein BLAST

BLASTP programs search protein databases using a protein query. [more...](#)

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#) [Clear](#)

Query subrange [?](#)

From

To

>unnamed protein product
GLGCGIGKMDMKACGLLVALGFVASCLAERDCTVSNFPVMQNFDRYRYTGTW
YAMAKKDPEGLFLLDNIV
AKFTVDDYGRMLATAKGRVVIFGGFQLCADMVGIFTDDDDPAKFVMQYRGLLAY

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No file chosen [?](#)

Job Title

Protein Sequence

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database

Non-redundant protein sequences (nr) [?](#)

Organism

Optional

Ambystoma tigrinum (taxid:8305)

☐ exclude [Add organism](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [?](#)

Exclude

Optional

☐ Models (XM/XP)
☐ Non-redundant RefSeq proteins (WP)
☐ Uncultured/environmental sample sequences

Program Selection

Algorithm

☐ Quick BLASTP (Accelerated protein-protein BLAST)
☒ blastp (protein-protein BLAST)
☐ PSI-BLAST (Position-Specific Iterated BLAST)
☐ PHI-BLAST (Pattern Hit Initiated BLAST)
☐ DELTA-BLAST (Domain Enhanced Lookup Time Accelerated BLAST)

Choose a BLAST algorithm [?](#)

BLAST

Search database nr using **Blastp** (protein-protein BLAST)
☐ Show results in a new window

feedback

 Your search is limited to records include: **Ambystoma tigrinum (taxid:8305)**

Job Title

Protein Sequence

RID

[Z9JHAX4U014](#)

Search expires on 05-02 23:43 pm

[Download All](#) ▾

Program

 [Citation](#) ▾

Database

nr [See details](#) ▾

Query ID

lcl|Query_7696226

Description

unnamed protein product

Molecule type

amino acid

Query Length

208

Other reports



Filter Results

Percent
Identity

to

E value

to

Query
Coverage

to

Filter

Reset



No significant similarity found. For reasons why, [click here](#)

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